

SAKET MAHESHWARI

Kasganj, Uttar Pradesh, India

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SUMMARY

AI/ML enthusiast and researcher specializing in deep learning applications with proven track record in biomedical signal processing, computer vision, and predictive analytics. Published author in IEEE with expertise in developing intelligent systems for real-world impact. Passionate about leveraging machine learning to solve complex technological challenges and mentoring peers in AI/ML domains.

EDUCATION

GLA University

Bachelor of Technology in Computer Science (AI & Machine Learning)

Nov 2022 – May 2026

Mathura, Uttar Pradesh

Mount Litera Zee School

Class 12 – Computer Programming

2019

EXPERIENCE

Research Intern

Indian Institute of Technology Jammu

Nov 2025 – June 2026

Jammu & Kashmir, India

- Developed deep learning models for biomedical signal analysis using TensorFlow and PyTorch, achieving 92% classification accuracy on ECG/EEG datasets
- Implemented signal preprocessing pipelines with wavelet transforms and filtering techniques, reducing noise artifacts by 40% and improving model performance
- Collaborated with research team on neural network architectures for real-time health monitoring, contributing to 4 research paper submissions

Research Internship in Sciences and Engineering

Indian Institute of Technology Jammu

June 2026 – Present

Jammu & Kashmir, India

- Designed a self-supervised lyrics intelligibility prediction system using Whisper encoder embeddings and Bidirectional LSTM networks
- Developed automated audio processing and pseudo-label generation pipelines for large-scale intelligibility assessment without manual annotations
- Achieved state-of-the-art prediction performance with RMSE 0.052 and Pearson correlation 0.89, resulting in a research paper submission in speech and audio signal processing

Machine Learning Researcher

Dr. B. R. Ambedkar National Institute of Technology Jalandhar

Jun 2025 – Sep 2025

Punjab, India

- Engineered transfer learning models for image classification using ResNet and VGG architectures, improving accuracy by 15% over baseline methods
- Designed AutoCAD-ML integration framework using computer vision to optimize building designs, reducing energy consumption predictions by 18%
- Published research on optimized image classification through transfer learning in IEEE conference proceedings

Machine Learning Intern

Feyn Labs

Jan 2024 – Mar 2024

Guwahati, Assam, India

- Built predictive maintenance system using Random Forest and LSTM models, achieving 87% failure prediction accuracy for industrial equipment
- Developed thermal energy forecasting model combining SARIMA and Random Forest, reducing prediction error by 23% compared to traditional methods
- Created data visualization dashboards using Python (Matplotlib, Seaborn) to communicate insights to non-technical stakeholders

PUBLICATIONS

Exploiting EEG-based Gamma-Band Time Frequency Feature in WaveNet Decoder Framework for High-Fidelity Speech Reconstruction

R. Buragohain, **S. Maheshwari** and K. Nathwani, “Exploiting EEG-based Gamma-Band Time Frequency Feature in WaveNet Decoder Framework for High-Fidelity Speech Reconstruction,” *Interspeech 2026*, *Accepted*.

HybridTrust: Fusing LP-GRU and Transformer Architectures for Enhanced Misinformation Detection in Social Networks

IEEE Xplore Link

S. Maheshwari, R. Chatterjee and S. Sinha, “HybridTrust: Fusing LP-GRU and Transformer Architectures for Enhanced Misinformation Detection in Social Networks,” *2025 First International Conference of Advances in Engineering and Computing Technologies for Sustainable Development (AECTSD)*, COIMBATORE, India, 2025, pp. 1–6, doi: 10.1109/AECTSD65988.2025.11411161.

Reducing Aircraft Downtime with Predictive Maintenance: A Data-Driven Approach for Airbus A320

[IEEE Xplore Link](#)

S. Maheshwari, S. Sinha and R. Chatterjee, "Reducing Aircraft Downtime with Predictive Maintenance: A Data-Driven Approach for Airbus A320," *2025 International Conference on Emerging Techniques in Computational Intelligence (ICETCI)*, Hyderabad, India, 2025, pp. 1–9, doi: 10.1109/ICETCI67340.2025.11257885.

DeepNeuroScan: Advanced CNN Architecture for Precise Brain Tumor Identification in Medical Imaging

[IEEE Xplore Link](#)

S. Maheshwari, R. Chatterjee and M. Agrawal, "DeepNeuroScan: Advanced CNN Architecture for Precise Brain Tumor Identification in Medical Imaging," *2025 IEEE International Conference on Computer, Electronics, Electrical Engineering & their Applications (IC2E3)*, Srinagar Garhwal, India, 2025, pp. 1–6, doi: 10.1109/IC2E365635.2025.11167740.

Forecasting Thermal Energy Demand: A Multimodal Approach Using Random Forest and SARIMA Models

[IEEE Xplore Link](#)

S. Maheshwari, M. Agrawal and S. Sinha, "Forecasting Thermal Energy Demand: A Multimodal Approach Using Random Forest and SARIMA Models," *2025 IEEE International Conference on Computer, Electronics, Electrical Engineering & their Applications (IC2E3)*, Srinagar Garhwal, India, 2025, pp. 1–5, doi: 10.1109/IC2E365635.2025.11167471.

Improved Diabetes Prediction Through Multimodeling: Merging Diverse Machine Learning Models for Superior Performance

[IEEE Xplore Link](#)

S. Maheshwari, "Improved Diabetes Prediction Through Multimodeling: Merging Diverse Machine Learning Models for Superior Performance," *2025 International Conference on Emerging Trends in Industry 4.0 Technologies (ICETI4T)*, Navi Mumbai, India, 2025, pp. 1–7, doi: 10.1109/ICETI4T63625.2025.11132156.

Enhancing Network Security through Real-Time Anomaly Detection in Network Traffic

[IEEE Xplore Link](#)

P. K. Jha, **S. Maheshwari**, R. Kumar and B. Kumar, "Enhancing Network Security through Real-Time Anomaly Detection in Network Traffic," *2025 International Conference on Intelligent and Cloud Computing (ICoICC)*, Bhubaneswar, India, 2025, pp. 1–6, doi: 10.1109/ICoICC64033.2025.11051998.

Predicting Telecom Customer Churn with Combined Machine Learning Models

[IEEE Xplore Link](#)

S. Sinha, **S. Maheshwari**, R. Chatterjee, P. Sharma, S. Saxena and H. Rajoria, "Predicting Telecom Customer Churn with Combined Machine Learning Models," *2025 IEEE International Conference on Emerging Trends in Computing and Communication (ETCOM)*, Mangalore, India, 2025, pp. 1–7, doi: 10.1109/ETCOM66606.2025.11437063.

Unlocking Dev Nagri: OCR with CNN

[SCRS Link](#)

Saket Maheshwari, Sayantan Sinha and Sachin Upadhyay, "Unlocking Dev Nagri: OCR with CNN", In: Mukesh Saraswat and Rajani Kumari (eds), *Applied Intelligence and Computing*, SCRS, India, 2024, pp. 157-167. <https://doi.org/10.56155/978-81-955020-9-7-17>

PROJECTS

Unlocking Deva Nagri: OCR with CNN | *Python, TensorFlow, OpenCV, CNN*

2024

- Co-authored OCR system for Devanagari script using convolutional neural networks, achieving 94% character recognition accuracy
- Implemented data augmentation techniques to expand training dataset by 300%, improving model generalization across font variations

Autism Detection System | *Python, scikit-learn, Machine Learning*

2024

- Developed ML classification models for early autism detection using behavioral data, achieving 89% diagnostic accuracy
- Applied feature engineering and hyperparameter tuning to optimize model performance, reducing false negatives by 25%

Urban Infrastructure Growth Prediction | *Python, Satellite Imagery, Deep Learning*

2024

- Created deep learning model to analyze satellite imagery for urban development patterns, processing 10,000+ images
- Implemented semantic segmentation using U-Net architecture to identify infrastructure growth with 86% IoU score

TECHNICAL SKILLS

Programming Languages: Python, C++, HTML, CSS, JavaScript, SQL

AI/ML Frameworks: TensorFlow, PyTorch, Keras, scikit-learn, OpenCV

Soft Skills: Research, Technical Writing, Mentoring, Team Collaboration

REFERENCES

Dr. Karan Nathwani

Associate Professor

Department of Electrical Engineering

Indian Institute of Technology Jammu

Email: karan.nathwani@iitjammu.ac.in

Dr. Sayantan Sinha

Technical Trainer

Department of Electrical Engineering

GLA University Mathura

Email: sayantan.sinha@gla.ac.in